

MQST Ensemble QC Laboratory

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Lab 3 - Quantum Gates

Winter quarter 2025

1 Introduction

In this set of labs, we will begin working with the carbon-13 labeled chloroform system as a two-qubit system. You will learn to describe single and two-qubit gates in terms of product operators and make the connection to the various calibrated NMR pulses used in the previous labs. Using calibrated single-qubit quantum gates, you will build up the pulse sequence that implements a two-qubit entangling gate, the CNOT. You will then learn about various techniques to prepare ‘initialized’ states, which for room-temperature liquid state NMR are known as ‘pseudo-pure-states’. We will implement one of these techniques and time-permitting perform (full or partial) quantum state tomography to verify our preparations.

1.1 Assigned Reading

- J. A. Jones, Quantum Computing with NMR, Progress in Nuclear Magnetic Resonance Spectroscopy 59, 91 (2011).

This is an excellent review of much of what we are doing. Sec. 2 is a brief review of what you have already learned. Sec's 3 and 4 are highly relevant to this lab.

- E. Knill, I. Chuang, and R. Laflamme, Effective Pure States for Bulk Quantum Computation, Phys. Rev. A 57, 3348 (1998).

This is work summarized in Jones Sec. 4.4 Temporal averaging. This is one of the approaches we will implement for state 'initialization'.

- Lu, D., Brodutch, A., Park, J., Katiyar, H., Jochym-O'Connor, T. and Laflamme, R., (2016). NMR quantum information processing. From Electron Spin Resonance (ESR) Based Quantum Computing, pp.193-226.

This is a book chapter on the implementation of quantum information processing using NMR. It provides a comprehensive description of the steps that we are taking in this lab.

Additional seminal papers worth a look :

- D. G. Cory, A. F. Fahmy, and T. F. Havel, Ensemble Quantum Computing by NMR Spectroscopy, PNAS 94, 1634 (1997).
- N. A. Gershenfeld and I. L. Chuang, Bulk Spin-Resonance Quantum Computation, Science 275, 350 (1997).
- D. G. Cory, M. D. Price, and T. F. Havel, Nuclear Magnetic Resonance Spectroscopy: An Experimentally Accessible Paradigm for Quantum Computing, Physica D: Nonlinear Phenomena 120, 82 (1998).

2 Prelab

Please do the assigned reading and complete the following problems :

1. Preparatory Question 1 in "Quantum Information Processing with NMR". See the file MIT-QCwithNMRlab.pdf.
2. The thermal state of chloroform.
 - a) Evaluate the (thermal) equilibrium density matrix, ρ_{eq} , for our two-qubit system encoded in the nuclear spins of carbon-13 labeled chloroform. Assume room temperature 20 °C, a magnetic field of exactly 1.5 T, and your measured chemical shifts and J coupling from lab 2. Please include enough significant digits so that the J coupling is accounted for. It may be quite a few...
 - b) Express ρ_{eq} in terms of product operators, and only up to a few significant digits. Compare with Eq. 4 of Knill et al., PRA 57, 3348 (1998).

3. Following Knill et al.'s procedure for 'Exhaustive Averaging' which we now call 'Temporal averaging', evaluate p , the 'excess probability' of the achievable pure state, starting from the chloroform thermal state evaluated in the previous problem. Hint: you can do most of this by inspection and substitution of our specific and higher precision values.
4. The controlled-NOT (CNOT) gate plays a special but not unique role as an archetypal two-qubit entangling gate. In the language we have been using this quarter, it is better referred to as a controlled-X gate. Jones Sec. 3.3 has a nice summary, including the relationship with a controlled-Z plus some Hadamards that you should already be familiar with.
 - a) Work out (don't just write down) the product operator expansion of CNOT, i.e. demonstrate that the unique product operator expansion of CNOT evaluates to the expected matrix representation of the gate.
 - b) Implement the gate sequence expressed by this product operator expansion in Python/QuTiP and demonstrate its behavior on a variety of relevant and interesting initial states. In the lab, you will take this and implement it as a pulse sequence and explore some approximate (compiled) forms.

3 Lab Procedure

Read the material and follow the instructions in the Python notebook *NMR_Lab3.ipynb* found on BruinLearn and on the Magritek systems.